GLIMPSE: GLOBAL–LOCAL INDEXING WITH MULTI-PRECISION SPECTRAL ENCODING FOR EXPLORABLE VIDEO WORLD MODELS

Zhufeng Xu[1,2] Henglei Lv[1,2] Bailin Deng[3] Yikang Ding[4]
Xiaoqiang Liu[4] Guoxin Zhang[4] Hongbo Fu[5] Lin Gao[1,2,*]

[1]Institute of Computing Technology, Chinese Academy of Sciences
[2]University of Chinese Academy of Sciences [3]Cardiff University
[4]Kling AI [*]Corresponding author: Lin Gao
[5]The Hong Kong University of Science and Technology



Figure 1: **Camera-controlled scene exploration with GLIMPSE.** From top to bottom: forward motion, left/right observation followed by backward motion, and free exploration over 81, 241, and 721 frames. Each row shows the requested camera trajectory on the left and five generated frames in temporal order. Insets highlight the corresponding camera pose, and colors indicate progression along the trajectory.

ABSTRACT

A camera-controlled video world model must keep a scene consistent when the camera returns to it. Memories that render stored features into the target frame can turn depth and pose errors into ghosting or duplicated structure. Positional transport avoids that resampling by relocating a historical patch's address rather than its content, but assigns full positional precision even to inaccurate projections. We present GLIMPSE, a global–local memory interface with band-wise multi-precision coordinates and geometry-conditioned spectral attenuation to reduce sensitivity to inaccurate projected addresses. Local patch memory supplies view-specific appearance, while recurrent global memory provides scene context. Footprint-guided retrieval selects historical patches under a fixed local read budget. Under matched retrieval and read budgets, GLIMPSE improves geometric correspondence and photometric consistency over deterministic positional transport.

1 INTRODUCTION

An explorable video world model generates the next video window along a requested camera trajectory. During extended exploration it must also remain self-consistent: surfaces should retain their appearance when the camera returns to them. Revisiting previously observed regions requires retrieving historical appearance and addressing it in the requested view. Yet geometric estimates are imperfect, making the reliability of these positional addresses critical to consistent generation.

Prior work follows three main approaches. Frame- and view-retrieval memories retain selected historical observations (Yu et al., 2025a; Xiao et al., 2025; Li et al., 2025b). A limited set of views

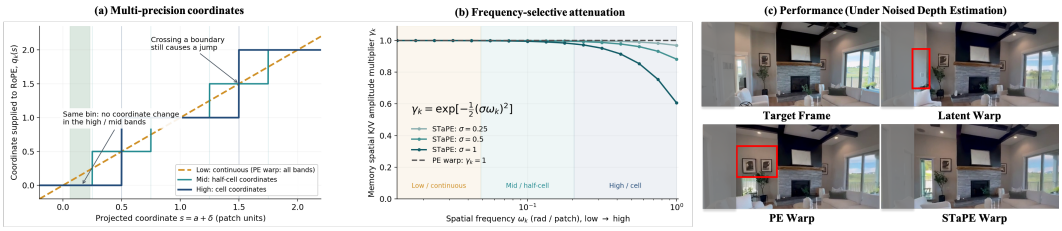


Figure 2: **Motivation for robust positional addressing.** *Left:* the coordinate supplied to RoPE against the continuous projected coordinate, where the high and mid bands hold a cell and a half-cell value, so a sub-cell error inside one bin changes nothing, while the low bands keep the continuous offset. *Middle:* the spectral multiplier γ_k against frequency, where PE warp keeps every band at full amplitude, whereas Spatio-Temporal Aligned Positional Encoding (STaPE) attenuates the highest frequencies first, more strongly as σ grows. *Right*, in reading order: the reference view, latent value warping, PE warping, and STaPE under the same four-patch address perturbation. Value warping drops the doorway and PE warping duplicates wall art, whereas STaPE preserves both.

can preserve scene identity but may omit details needed by a returning viewpoint. Value-warping memories instead reconstruct geometry and render stored features onto the target grid (Wu et al., 2025; Wang et al., 2026a). This provides explicit geometric alignment, but resolves estimated depth and pose into pixel positions, so reprojection errors can propagate into generation as ghosting or duplicated structure. Positional transport avoids this resampling by leaving each historical patch exactly as it was observed and relocating only its address within attention (Xu et al., 2026b; Yu et al., 2026). Each token remains associated with its original patch, without resampling historical features onto the target grid.

Positional transport preserves historical appearance, but relies on estimated geometry to locate it in the target view. Errors in depth or camera pose can misplace these positional addresses, with high-frequency rotary bands being particularly sensitive to small coordinate errors. Standard positional transport does not account for this varying geometric reliability. We therefore adapt the precision and frequency content of each address to the reliability of its underlying geometry, preserving fine positional detail when reliable and reducing sensitivity to uncertain projections otherwise (Figure 2).

We introduce **GLIMPSE**, **G**lobal–**L**ocal **I**ndexing with **M**ulti-**P**recision **S**pectral **E**ncoding, a global–local memory framework for explorable video generation (Figure 1). In our framework, a frozen recurrent memory provides global context, while a local pointer memory retrieves original historical patches through projected footprint coverage under a fixed per-frame read budget. The two memories are accessed through separately gated readouts. To address retrieved patches in the requested view, we introduce Spatio-Temporal Aligned Positional Encoding (STaPE), which represents projected locations at band-specific coordinate precisions and regulates their spectral strength using geometric reliability cues. This design reduces sensitivity to projection errors without resampling historical appearance.

Our contributions are threefold:

- We propose a **geometry-addressed global–local memory interface** that separates scene-level recurrent context from view-specific appearance. Footprint-guided retrieval selects original historical patches under a fixed local read budget. Separate gated readouts combine the two memories without making global context compete with local evidence in the same attention normalization.
- We introduce **multi-precision positional addressing**, assigning cell, half-cell, and continuous coordinates to different rotary frequency bands. Fine-frequency addresses remain unchanged under perturbations within a bin, while continuous lower-frequency coordinates retain sub-cell placement without adding memory tokens.
- We develop **geometry-conditioned spectral attenuation**, using depth smoothness and temporal lag to regulate spatial frequency components of memory keys and values. Together with multi-precision addressing, it controls the precision and strength of transported addresses. Matched ablations establish their complementary benefits over deterministic positional transport in geometric correspondence and photometric consistency.

2 Related Work

2.1 Interactive Video World Models

Interactive video world models generate observations in response to user actions while keeping the environment coherent across interaction steps. In camera-directed systems, the action specifies a trajectory and the observation is the next video window. InSpatio-World develops this autoregressive trajectory-control interface, while AlayaWorld, Wonder, DreamX-World, and Infinite-World target long-running playable or general-purpose environments (InSpatio Team et al., 2026; AlayaWorld Team et al., 2026; Xu et al., 2026a; DreamX Team et al., 2026; Wu et al., 2026a). GLIMPSE focuses on retaining historical appearance and retrieving it for a new camera query. Vid2World adapts pretrained video diffusion models to causal interactive generation (Huang et al., 2026), and MiniWorld instead studies a reproducible streaming training recipe from scratch (Zhao et al., 2026). GLIMPSE studies the memory interface within a pretrained video generator.

2.2 Camera Control and Positional Alignment

Camera-controllable generation conditions synthesis on an explicit viewpoint trajectory. CameraCtrl injects camera-ray conditioning (He et al., 2024), whereas TrajectoryCrafter jointly conditions on point-cloud renders and source video through a dual-stream diffusion model (Yu et al., 2025b). GEN3C conditions generation on a point-cloud cache rendered at the requested camera (Ren et al., 2025). UCPE combines Relative Ray Encoding of 6-DoF pose, intrinsics, and lens distortion with Absolute Orientation Encoding of pitch and roll through a lightweight spatial-attention adapter (Zhang et al., 2026). PRoPE expresses relative camera geometry as projective positional encoding inside attention (Li et al., 2025a). CameraSquad combines decoupled content and camera control with dual-mode cross-view attention for consistent parallel multi-trajectory generation (Xu et al., 2026c). These encode geometry to *control* generated target tokens. UCM applies the same VAE and patchification to historical and target latents, then warps positional encoding in time, assigning target space-time coordinates to clean historical tokens while leaving their content unchanged (Xu et al., 2026b). Integrated positional encoding attenuates Fourier features according to spatial uncertainty (Barron et al., 2021), and our work instead considers geometry-conditioned positional addressing for retrieved video memory.

2.3 Persistent and Spatial Memory

Recent-context methods improve how a finite frame history is packed or weighted (Zhang et al., 2025; Song et al., 2025; Shen et al., 2026). Persistent-memory methods retain information beyond that window: Context as Memory and WorldMem retrieve stored frames, VMem retrieves views through a surfel index of the queried surface, and SPMem combines working, spatial, and episodic stores (Yu et al., 2025a; Xiao et al., 2025; Li et al., 2025b; Wu et al., 2025). CUT3R maintains a recurrent state for streaming 3D perception and MeMix updates that state selectively to reduce forgetting (Wang et al., 2025b; Dong et al., 2026). GLIMPSE uses these frozen components for its global branch. Long-context state-space video world models combine blockwise state-space scanning with local attention (Po et al., 2025). GIM-World compresses history into learned geometry-aware implicit memory and prunes observations (Wei et al., 2026). In contrast to this compressed history, LPM retains the provenance of individual source patches. WorldPlay reconstructs context from past frames and temporally reframes old observations to maintain geometric consistency (Sun et al., 2025).

A further line of work makes spatial memory directly addressable. Mirage persists latent features at world points and renders their values onto a target grid, MosaicMem indexes localized patches and aligns them through warped RoPE or warped latents, and SPMem renders explicit spatial-memory values in the target view alongside sparse episodic frames (Wang et al., 2026a; Yu et al., 2026; Wu et al., 2025). The value-warping variants commit appearance to a target raster before attention, whereas positional-warping variants preserve separate source features. AnchorWeave retrieves local geometric memories by trajectory coverage and renders them as anchor conditions (Wang et al., 2026b), and WorldTrace keeps a compressed cache addressable by assigning summary slots virtual positions (Wu et al., 2026b). Concurrently, ReWorld bounds long-horizon memory with a pose-indexed landmark bank under a fixed KV budget (Chen et al., 2026). LPM uses the positional-warping

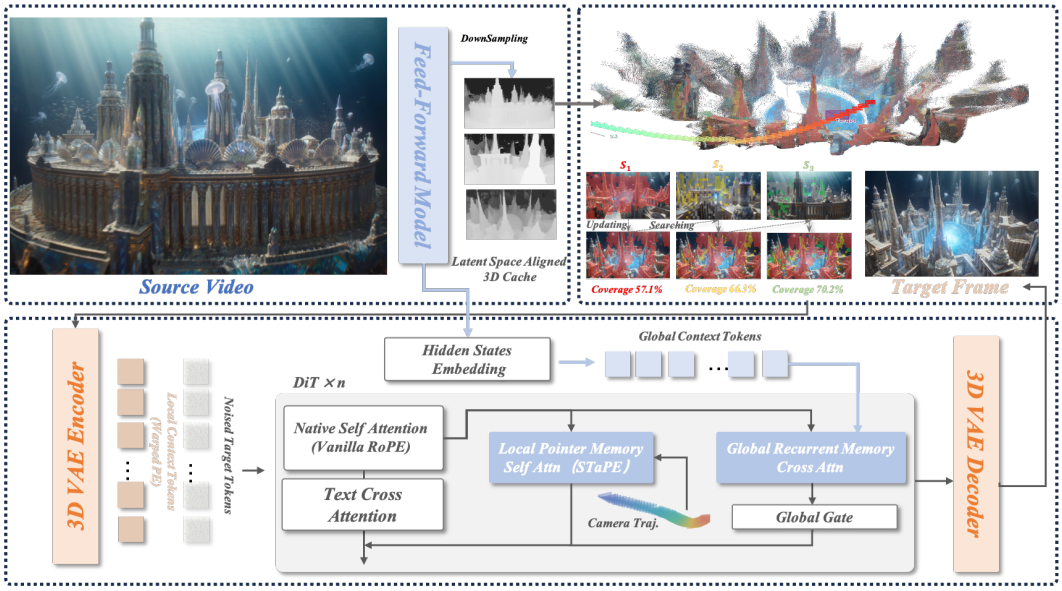


Figure 3: Overview of GLIMPSE. Geometry-guided greedy retrieval selects local context from the source history. STaPE aligns the positions of retrieved latent tokens with the requested target view, while recurrent global context provides complementary scene memory to the video generation model.

route with a fixed per-target-frame token budget: footprint coverage selects views, and a separate allocator gathers the original patches that enter attention.

3 METHOD

3.1 FRAMEWORK OVERVIEW

Figure 3 illustrates the overall framework. At exploration step t the user specifies a camera action $a_t = (T_t^q, K_t^q)$: the requested target pose and intrinsics. Let x_t^{src} denote the current source observation and $S_t = (G_t, L_t)$ the global and local memories. The local memory also stores camera and time indices. Generation and memory updates follow

$$a_t = (T_t^q, K_t^q), \qquad o_{t+1} \sim p_\theta(o \mid x_t^{\text{src}}, a_t, G_t, L_t), \qquad S_{t+1} = \mathcal{U}(S_t, o_{t+1}, a_t). \tag{1}$$

Each update advances GRM and appends observations and geometry to LPM. Memory read length is fixed, but storage, indexing, and geometry estimation grow with history. Appendix B.5 profiles geometry, pointer-cache, and denoising costs across horizons.

3.2 GLOBAL–LOCAL MEMORY

The GRM G_t is a frozen CUT3R (Wang et al., 2025b) recurrent state built by processing selected observations in chronological order, using the parameter-free MeMix update to combine retained state tokens with each new observation. It provides a fixed-size summary of the scene and its history. The LPM is a geometry-indexed interface over observation history,

$$I_t = \{(\mathbf{X}_j, \mathbf{d}_j, \iota_j, c_j, \alpha_j)\}_{j \in \mathcal{H}_t}, \quad R_{t,f}^q = \mathcal{Q}(I_t; T_{q,f}, K_{q,f}), \quad |R_{t,f}^q| \le M_f, \tag{2}$$

where $\mathbf{X}_j$ and $\mathbf{d}_j$ denote the world position and viewing ray, ι_j identifies the original observation and latent patch, and c_j and α_j record confidence and temporal lag. We rebuild I_t from the available history geometry at each query and limit each target frame's read to one latent frame's worth of tokens. The DiT receives the original patch tokens selected by $R_{t,f}^q$, not the index itself. We rank candidate views by projected footprint coverage, then select original patch tokens from these views based on their projected centers.

LPM tokens have explicit target-view coordinates, and GRM tokens are addressed by content. We read GRM through a separately gated cross-attention, so global tokens do not compete with local evidence for attention weight.

Retrieval-to-read procedure. Let $H_p \times W_p$ be the shared patch-embedding grid and $M = H_p W_p$. For each target frame q, candidate views are jointly ranked by camera proximity and temporal lag, prioritizing source observations over generated history. Using the occupancy masks F_i of projected patch quadrilaterals, greedy selection adds the view covering the most uncovered cells at each round. In selected-view order the allocator keeps the first positive-depth, in-bounds projected center for each target cell and fills the remaining capacity up to M with unused original tokens. These fillers may be redundant or outside the target frustum. The allocator preserves individual token features and applies no target-depth test. Table 13 compares this allocator with center-hit and causal z-buffer controls under a matched token budget.

At each denoising step, history and main-stream latents share the DiT patch embedding. Each backbone layer updates pointer tokens at timestep zero using source-coordinate native RoPE and a $\tanh(g_\ell)$ gate, then processes the main stream. Target queries read the associated pointer K/V through STaPE and GRM through a separate softmax. The two outputs form the conditioning residual (Equation 14). Appendix A provides the implementation settings and retrieval details.

3.3 Spatio-Temporal Aligned Positional Encoding

STaPE places a historical patch in the continuous spatial coordinate system of its associated target frame. For latent-cell center $\mathbf{x}_h$, historical depth D_h, historical pose T_h, and target camera (T_q, K_q),

$$(u_h^q, v_h^q) = \mathrm{project}(K_q, K_h, T_q, T_h, D_h, \mathbf{x}_h), \qquad \phi(h) = q. \tag{3}$$

Here T_h, T_q are camera-to-world poses in the same DA3 scene gauge, and K_h, K_q are intrinsics at the coordinate resolution used in the projection. For patch-grid center $\mathbf{x}_h = (c + \frac{1}{2}, r + \frac{1}{2}, 1)^\top$, the implementation unprojects $\mathbf{X}_h = R_h(d_h K_h^{-1}\mathbf{x}_h) + \mathbf{t}_h$ and transforms it to the target as $\mathbf{X}_q = R_q^\top(\mathbf{X}_h - \mathbf{t}_q)$. Homogeneous division of $K_q\mathbf{X}_q$ gives image coordinates, which are scaled to target patch-grid units and shifted by $-\frac{1}{2}$ so integer coordinates denote patch centers. Appendix A.2 specifies depth resizing and the separate handling of projected support and budget fillers. The projection retains a continuous center estimate, and the band-dependent coordinate rule below then selects its encoding precision. Membership $\phi(h) = q$ supplies temporal alignment through the frame-local adapter mask. No separate temporal rotary phase is introduced in this adapter.

Target and retrieved history latents pass through the same DiT patch embedding E, adapter normalization and projection A, and key/value projections. The pointer refresh reuses native backbone weights, whereas each layer's adapter has separate normalization and projections initialized from the corresponding DiT block. Appendix A.2 gives the execution order and the detailed shadow update and residual equations. Let $\mathbf{h}_i^0 = E(\mathbf{z}_i)$ and let $\mathbf{h}_i^\ell$ denote the hidden feature supplied to the local read at layer ℓ after the above processing. The content tensors before positional encoding are

$$\begin{aligned} \mathbf{f}_i &= A(\mathrm{Norm}(\mathbf{h}_i^\ell)), & Q_q^{\mathrm{pre}} &= W_Q\,\mathrm{Norm}_Q(\mathbf{f}_q), \\ K_i^{\mathrm{pre}} &= W_K\,\mathrm{Norm}_K(\mathbf{f}_i), & V_i^{\mathrm{pre}} &= W_V\mathbf{f}_i. \end{aligned} \tag{4}$$

Here Norm, Norm_Q, and Norm_K are the adapter's learned LayerNorms, applied before head reshaping. Let P_q and P_h be the target and historical token's source-camera projective matrices, incorporating intrinsics in a shared scene gauge. PRoPE partitions each head into a $D/2$ camera block and two $D/4$ spatial blocks (Li et al., 2025a). Appendix A.2 defines the corresponding Q, K/V, and O basis matrices. Within either spatial block, pair k consists of channels $(k, k+m)$, $m = D/8$, and

$$R_k(a) = \begin{bmatrix} \cos(\omega_k a) & \sin(\omega_k a) \\ -\sin(\omega_k a) & \cos(\omega_k a) \end{bmatrix}, \qquad \omega_k = 100^{-k/m}. \tag{5}$$

The adapter and native DiT backbone use their respective frequency schedules.

Geometry-conditioned spectral regularization. We attenuate both channels of each spatial rotary pair by $\gamma_k = \exp[-(\sigma\omega_k)^2/2]$, motivated by Gaussian perturbations of continuous coordinates (Appendix A.1). Collecting these factors in $\Gamma(\sigma)$ gives the operator

$$\begin{aligned} D_\sigma &= \mathrm{diag}\big(I_{D/2}, \Gamma(\sigma), \Gamma(\sigma)\big)\,, \\ \mathcal{P}_{h,\sigma}^{KV}(u,v) &= D_\sigma B_h^{KV}(u,v). \end{aligned} \tag{6}$$

Memory K/V use P_h^{-1} with target-projected spatial coordinates, while attenuation leaves camera components and target queries unchanged. Let $a_q = (u_q, v_q)$ and let Q_q denote the target query transformed by the standard PRoPE operator $B_q^Q(a_q)$. For local memory $H_q = \{h : \phi(h) = q\}$, the positional transformation and memory attention are defined as follows:

$$\begin{aligned}(K_h^{\mathrm{pos}}, V_h^{\mathrm{pos}}) &= \mathcal{P}_{h,\sigma_h}^{KV}(u_h^q, v_h^q)(K_h^{\mathrm{pre}}, V_h^{\mathrm{pre}}),\\ Y_q &= B_q^O(a_q)\,\mathrm{Attn}\left(Q_q, K_{H_q}^{\mathrm{pos}}, V_{H_q}^{\mathrm{pos}}\right).\end{aligned} \tag{7}$$

Superscripts pre and pos denote features before and after positional transformation. We determine each token's attenuation width from local depth smoothness and temporal lag:

$$\begin{aligned}c_h &= \mathrm{clip}_{[0,1]} \exp\left[-4\,\frac{|\Delta_x d_h| + |\Delta_y d_h|}{\max(d_h, 10^{-3})}\right],\\ \alpha_h &= 1 - \bar{t}_h,\\ \sigma_h &= \lambda_c[1 - c_h]_+ + \lambda_a[\alpha_h]_+.\end{aligned} \tag{8}$$

Here $\bar{t}_h \in [0, 1]$ is the normalized observation time. Larger widths suppress high spatial frequencies without changing the projected coordinate. At $\sigma_h = 0$, attenuation is disabled, while the multi-precision coordinate rule remains active.

Multi-precision coordinates. For pair index k, let $b_1 = \lceil m/3 \rceil$ and $b_2 = \lceil 2m/3 \rceil$. We replace the coordinate in each spatial rotation by

$$q_k(a) = \begin{cases}\mathrm{round}_{\mathrm{even}}(a), & k < b_1,\\ \frac{1}{2}\,\mathrm{round}_{\mathrm{even}}(2a), & b_1 \le k < b_2,\\ a, & k \ge b_2.\end{cases} \tag{9}$$

Here $\mathrm{round}_{\mathrm{even}}$ rounds to the nearest integer, ties to even. High-frequency pairs use cell coordinates, middle pairs use half-cell coordinates, and low-frequency pairs retain continuous offsets. In the quantized bands, spatial phases remain unchanged within each bin and change discontinuously across bin boundaries. For this variant, $R_k(a)$ in Equation 5 is replaced by $R_k(q_k(a))$ and the attenuation in Equation 6 is unchanged. Source-token identity is preserved throughout: the operator transforms feature channels without spatially resampling or averaging appearance between tokens.

3.4 Training and Data Construction

We train on a mixture of SpatialVID (Wang et al., 2025a), RealCam-Vid (Zheng et al., 2025), ReCamMaster (Bai et al., 2025), and Context-as-Memory (Yu et al., 2025a). Cached DA3 (Lin et al., 2025) pose, intrinsics, and depth are the only geometry source, and a cache miss skips the sample instead of invoking online estimation. Three augmentations stress the memory interface. *Future-as-memory* uses real post-target observations as training-only LPM donors with relabeled memory times. Inference uses only source observations and completed generated windows. *Synthetic revisitation* builds complementary old-view memory from multiple keyframes under a shared camera perturbation, and *fast motion* temporally subsamples the target at stride two or three while keeping the dense history. Sampling weights, offsets, probabilities, and eligibility rules are specified in Appendix B. Table 7 isolates the augmentations through separately trained variants, including causal-only donors, evaluated under the same causal inference protocol.

Training initializes the dense DiT from Wan2.2-TI2V-5B, runs at 256×448, and continues at 352×640. Only the target latent is noised. We parameterize the interpolation by the noise level $s = \sigma_j$, selected from the scheduler, and $t_j = 1000s$ is the timestep supplied to the DiT. For $\epsilon \sim \mathcal{N}(0, I)$,

$$\begin{aligned}Z_s &= (1 - s)Z_0 + s\epsilon,\\ \mathcal{L}_{\mathrm{FM}} &= \mathbb{E}_{j,\epsilon}\left[w_j \left\|v_\theta(Z_s, t_j; x^{\mathrm{src}}, a, G, L) - (\epsilon - Z_0)\right\|_2^2\right].\end{aligned} \tag{10}$$

Thus v_θ predicts $\partial Z_s/\partial s = \epsilon - Z_0$. Inference integrates in s using the Euler update $Z_{s'} = Z_s + (s' - s)v_\theta(Z_s, 1000s; \cdot)$, ending at $s' = 0$. Appendix B.1 gives the shifted nodes and loss weights.

After each window, DA3 re-estimates source and generated-history geometry with the source anchor fixed. Subsequent commands, intrinsics, and camera history are rebased onto this estimated trajectory.

Table 1: Camera-control and video-quality metrics. Rotation and direction errors are in degrees. Bold marks the best score.

Metric	GLIMPSE (ours)	UCM	SPMem	GEN3C	VMem	ReWorld
Mean rot.↓	**0.77**	0.88	5.56	1.29	7.87	2.94
Endpoint rot.↓	**1.28**	1.48	5.68	2.95	9.50	4.78
Trans. NRMSE↓	0.102	**0.059**	0.318	0.342	0.525	0.424
Trans. dir.↓	**2.14**	5.54	9.20	9.31	12.42	8.03
CLIP-T↑	**30.445**	29.486	28.641	29.041	26.357	30.012
CLIP-F↑	**99.123**	99.006	98.808	98.912	97.509	99.101
FID↓	**62.218**	62.387	65.487	63.874	89.887	62.307
FVD↓	863.838	877.592	940.247	903.112	1044.543	**854.19**

Table 2: WorldScore components for the camera-control experiment. Scores are normalized to 0–100 and higher is better, including transformed error scores. The mean weights all nine components equally. Bold marks the best score.

Metric	GLIMPSE (ours)	UCM	SPMem	GEN3C	VMem	ReWorld
Camera	**91.032**	90.245	81.273	82.510	81.294	84.740
Text	**88.983**	87.579	87.509	83.072	71.927	88.703
3D reprojection	83.930	77.286	64.313	75.779	50.567	**85.149**
Flow AEPE	**89.752**	87.874	79.693	84.519	64.088	85.933
CLIP-IQA+	**35.958**	33.542	22.470	28.228	17.109	34.098
Aesthetic quality	**15.357**	13.081	11.010	13.875	11.920	13.882
Motion magnitude	34.026	33.702	33.805	**37.658**	28.592	34.132
Motion smoothness	**66.605**	62.055	65.267	61.899	44.810	66.227
Style consistency	94.380	**94.488**	89.305	93.297	85.011	92.012
Mean (9 metrics)	**66.669**	64.428	59.405	62.315	50.591	64.986

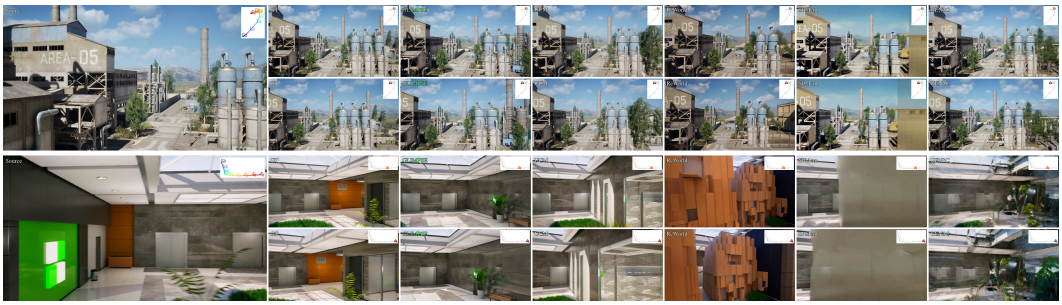


Figure 4: Camera-trajectory replay on two scenes. The enlarged source image is followed by GT, GLIMPSE, UCM, ReWorld, SPMem, and GEN3C (left to right), with two selected time steps per scene. Insets show the requested camera trajectory and current pose.

4 Experiments

4.1 Experimental Setup

GLIMPSE uses Wan2.2-TI2V-5B (Team Wan et al., 2025), cached DA3 geometry (Lin et al., 2025), and frozen CUT3R recurrent context (Wang et al., 2025b; Dong et al., 2026). Camera control uses trajectory errors, CLIP-T/F, FID, and FVD. WorldScore (Duan et al., 2025) evaluates quality and consistency, while Y-traj. also measures revisit correspondences and RGB agreement. We compare against UCM (Xu et al., 2026b), SPMem (Wu et al., 2025), GEN3C (Ren et al., 2025), VMem (Li et al., 2025b), and the concurrent ReWorld (Chen et al., 2026), following their public inference recipes. These comparisons evaluate complete systems, while matched internal ablations isolate the contribution of STaPE. SPMem denotes the spatial-memory video world model of Wu et al. (2025). Training settings, metric definitions, and comparison details are collected in Appendix B.2.

4.2 Camera-Control Evaluation

Table 1 summarizes camera diagnostics and video-quality metrics (Bai et al., 2025). GLIMPSE attains the lowest rotation errors and the lowest translation-direction error, together with the best CLIP-T, CLIP-F and FID. UCM achieves lower translation NRMSE, while ReWorld achieves lower FVD. Table 2 reports the WorldScore components on the same subset: GLIMPSE leads six of the nine and the nine-component mean, ReWorld ranks second overall and leads on 3D reprojection, while UCM leads on style consistency. Figure 4 shows qualitative replay comparisons.

4.3 Y-traj. Evaluation

We introduce the *Y-traj.* protocol to test whether a model can combine complementary observations in memory. Each sequence observes the left and right sides of a scene, then returns to the initial heading while moving backward to a wider view, so the final frame can contain regions seen at different times that no single earlier view covers. A trajectory that merely retraces its own path never poses that

Table 3: WorldScore components on 100 Y-traj. cases. Scores are normalized to 0–100 and higher is better, including transformed errors. The mean weights all ten components equally.

Metric	GLIMPSE (ours)	UCM	SPMem	GEN3C	VMem	ReWorld
CLIP-IQA+	**21.826**	14.161	11.075	14.631	12.113	20.922
Aesthetic quality	**32.323**	24.179	18.597	16.857	17.905	31.493
Text alignment	81.703	**82.122**	81.455	80.903	78.199	80.019
Style consistency	**88.531**	80.825	72.095	85.679	72.838	79.306
DINO consistency	83.390	80.912	77.655	80.453	73.990	**84.307**
CLIP consistency	**94.062**	93.842	93.105	93.367	92.231	93.065
Motion magnitude	**67.417**	66.050	57.270	59.862	52.451	64.114
Flow AEPE	**74.984**	74.644	67.875	73.798	63.090	74.578
Motion smoothness	33.685	30.847	**36.094**	30.357	30.917	30.281
Camera control	**87.596**	86.573	80.739	83.268	74.142	81.855
Ten-component mean	**66.552**	63.415	59.596	61.918	56.788	63.994

Table 4: Revisit correspondences on 100 Y-traj. cases. @1/@5 use the last one/five pairs per branch. Counts are in thousands (K) per pair, and RGB MAE uses $[0, 1]$ colors.

Metric	GLIMPSE (ours)	UCM	SPMem	GEN3C	VMem	ReWorld
Mat. Pix.@1 (K) ↑	**5.602**	5.307	2.592	4.223	2.138	4.799
Mat. Pix.@5 (K) ↑	**5.898**	5.335	2.621	4.298	2.733	4.296
Inliers@1 (K) ↑	**4.846**	4.338	2.149	3.722	1.430	3.704
Inliers@5 (K) ↑	**5.067**	4.373	2.156	3.791	1.826	3.717
Inlier rate@1 (%) ↑	**82.301**	75.752	80.048	77.810	72.885	72.991
Inlier rate@5 (%) ↑	**79.353**	78.922	76.440	77.133	72.813	72.744
RGB MAE@1 ↓	0.093	0.095	0.102	**0.079**	0.147	0.099
RGB MAE@5 ↓	0.091	0.097	0.102	**0.086**	0.152	0.099



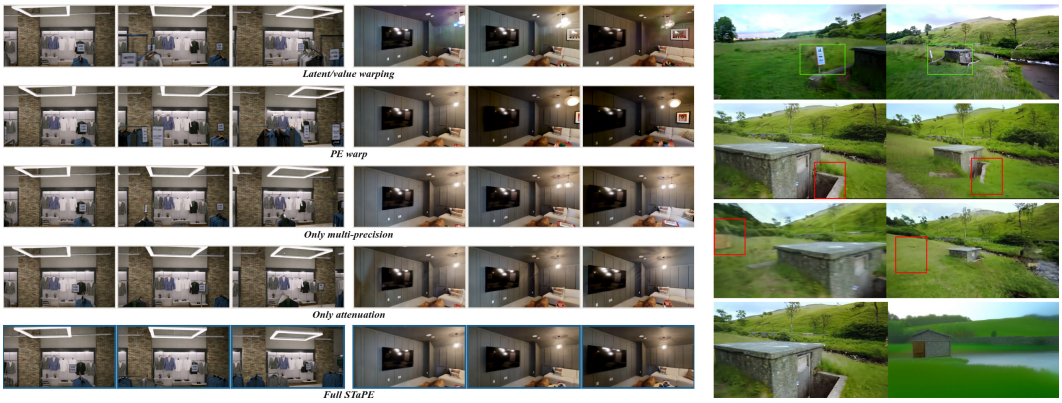
Figure 5: Y-traj. comparisons on house and street scenes. References and trajectories appear on the left, with left-yaw, right-yaw, and backward views shown top to bottom for each method. Red boxes highlight inconsistencies.

problem. Table 3 reports WorldScore results on 100 such cases. GLIMPSE attains the best value on seven of the ten components and the highest ten-component mean. The remaining three go to different systems: UCM on text alignment, ReWorld on DINO consistency, and SPMem on motion smoothness.

Revisit correspondences. We measure cross-view consistency by matching the earlier left/right observations to the final backward view (Table 4). GLIMPSE achieves the highest mean match and inlier counts, while GEN3C yields the lowest RGB MAE over accepted correspondences. Matching details are provided in Appendix B.2. Figure 5 highlights cross-view inconsistencies: UCM introduces a car and an additional street pole absent from the earlier views, while SPMem changes the house's appearance. UCM and GEN3C also exhibit spatial drift and local artifacts. GLIMPSE better preserves the scene layout and appearance in these examples. Additional results are provided in Appendix D.

4.4 Ablations

Tables 5–6 isolate STaPE and the memory branches. Additional diagnostics support frequency-selective, token-specific modulation over matched scalar suppression and permuted widths (Tables 8–9). Under identical address perturbations, STaPE retains more geometric inliers and lower RGB MAE than continuous PE transport (Table 10).



(a) STaPE components *(b) Global and local memory*

Figure 6: Qualitative ablations. **(a)** STaPE variants from Table 5 on two held-out scenes. **(b)** Global–local memory variants from Table 6 on a leave-and-return trajectory. Green and red boxes highlight preserved content and inconsistencies upon return, respectively.

Table 5: STaPE component ablation. Multi-precision coordinates and spectral attenuation are varied independently while footprint retrieval, local token budget, and GRM remain fixed. The latent/value-warping control resamples historical features onto the target grid and uses neither component. Both ablation tables use the same separate evaluation set with several trajectory types, including Y-traj. Rotation error is in degrees.

Configuration	Multi-precision	Spectral attenuation	Mat. Pix.@5 (K) ↑	Inliers@5 (K) ↑	Inlier rate@5 (%) ↑	RGB MAE@5 ↓	Mean rot. (°) ↓	Trans. NRMSE ↓	CLIP-T ↑	CLIP-F ↑
Latent/value warping	N/A	N/A	6.410	5.530	86.271	0.121	1.27	0.270	27.08	96.11
PE warp	Off	Off	6.407	5.690	88.002	0.135	1.30	0.213	29.314	97.43
Only multi-precision	On	Off	6.567	5.707	87.656	0.122	1.19	0.210	29.618	97.71
Only attenuation	Off	On	6.917	6.097	88.145	0.120	1.25	0.212	29.780	98.57
Full STaPE	On	On	7.153	6.480	90.587	0.109	1.16	0.204	30.089	99.13

Table 6: Global recurrent (GRM) and local pointer (LPM) memory ablation. Source, text, and camera conditions are retained. LPM uses full STaPE at a fixed read budget, with the protocol of Table 5.

Configuration	GRM	LPM	Mat. Pix.@5 (K) ↑	Inliers@5 (K) ↑	Inlier rate@5 (%) ↑	RGB MAE@5 ↓	Mean rot. (°) ↓	Trans. NRMSE ↓	CLIP-T ↑	CLIP-F ↑
No memory	Off	Off	5.393	4.693	87.021	0.183	1.27	0.316	29.771	97.50
Global only	On	Off	5.743	5.200	90.540	0.152	1.20	0.304	29.939	98.49
Local only	Off	On	6.637	6.007	90.507	0.133	1.26	0.298	30.012	97.99
Global + local	On	On	7.153	6.480	90.587	0.109	1.16	0.204	30.089	99.13

5 Conclusion and Limitations

We presented GLIMPSE, a global–local memory indexing framework with multi-precision spectral encoding for explorable video world models. Local pointer memory preserves view-specific historical evidence, while recurrent global memory supplies complementary scene context. At the core of this interface, STaPE addresses a challenge that persists even when historical content is preserved: inaccurate geometry can assign that content an overly precise positional address. By combining band-wise multi-precision coordinates with geometry-conditioned spectral modulation, STaPE improves memory reads without resampling historical appearance or increasing the local token budget. Camera-control and Y-trajectory experiments demonstrate improved cross-view correspondence and recovery of previously observed content. Matched component ablations support the complementary roles of the two addressing mechanisms and memory branches, while controlled address perturbations show reduced sensitivity to projection errors. Together, these results highlight that consistent exploration depends not only on retaining relevant history, but also on how its geometric addresses enter attention.

However, the current system focuses on camera-conditioned exploration and does not explicitly model object motion or use few-step distillation for real-time inference. Extending memory updates to track changing object states and distilling the generator to reduce denoising latency are natural next steps toward more responsive exploration and interaction.

REFERENCES

AlayaWorld Team, Kaipeng Zhang, Chuanhao Li, Yifan Zhan, Yongtao Ge, Yuanyang Yin, Jiaming Tan, Kang He, Liaoyuan Fan, Ruicong Liu, Xiaojie Xu, Xuangeng Chu, Zhen Li, Zhengyuan Lin, Zhixiang Wang, Zian Meng, and Zihui Gao. AlayaWorld: Long-horizon and playable video world generation. *arXiv preprint arXiv:2607.06291*, 2026. URL https://arxiv.org/abs/2607.06291.

Jianhong Bai, Menghan Xia, Xiao Fu, Xintao Wang, Lianrui Mu, Jinwen Cao, Zuozhu Liu, Haoji Hu, Xiang Bai, Pengfei Wan, and Di Zhang. ReCamMaster: Camera-controlled generative rendering from a single video. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 14834–14844, 2025. URL https://openaccess.thecvf.com/content/ICCV2025/html/Bai_ReCamMaster_Camera-Controlled_Generative_Rendering_from_A_Single_Video_ICCV_2025_paper.html.

Jonathan T. Barron, Ben Mildenhall, Matthew Tancik, Peter Hedman, Ricardo Martin-Brualla, and Pratul P. Srinivasan. Mip-nerf: A multiscale representation for anti-aliasing neural radiance fields. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021. URL https://jonbarron.info/mipnerf/.

Zhifei Chen, Luozhou Wang, Guibao Shen, Dongyu Yan, Shuai Yang, Tianshuo Xu, Yihua Du, Wei Wang, Tianyi Gui, Lianghua Huang, and Yingcong Chen. ReWorld: An interactive world model with long-horizon memory. *arXiv preprint arXiv:2608.23565*, 2026. URL https://arxiv.org/abs/2608.23565.

Jiacheng Dong, Huan Li, Sicheng Zhou, Wenhao Hu, Weili Xu, and Yan Wang. MeMix: Writing less, remembering more for streaming 3d reconstruction. *arXiv preprint arXiv:2603.15330*, 2026. URL https://arxiv.org/abs/2603.15330.

DreamX Team, Yancheng Bai, Rui Chen, Xiangxiang Chu, Rujing Dang, Hao Dou, Bingjie Gao, Qiwen Gu, Siyu Hong, Jiachen Lei, Geng Li, Jifan Li, Ruimin Lin, Qingfeng Shi, Bingze Song, Lei Sun, Jing Tang, Ruitian Tian, Jun Wang, Jiahong Wu, Pengfei Zhang, Shen Zhang, and Jiashu Zhu. DreamX-World 1.0: A general-purpose interactive world model. *arXiv preprint arXiv:2606.16993*, 2026. URL https://arxiv.org/abs/2606.16993.

Haoyi Duan, Hong-Xing Yu, Sirui Chen, Li Fei-Fei, and Jiajun Wu. WorldScore: A unified evaluation benchmark for world generation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 27713–27724, 2025. URL https://openaccess.thecvf.com/content/ICCV2025/html/Duan_WorldScore_A_Unified_Evaluation_Benchmark_for_World_Generation_ICCV_2025_paper.html.

Johan Edstedt, Ioannis Athanasiadis, Mårten Wadenbäck, and Michael Felsberg. DKM: Dense kernelized feature matching for geometry estimation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 17765–17775, 2023. URL https://openaccess.thecvf.com/content/CVPR2023/html/Edstedt_DKM_Dense_Kernelized_Feature_Matching_for_Geometry_Estimation_CVPR_2023_paper.html.

Hao He, Yinghao Xu, Yuwei Guo, Gordon Wetzstein, Bo Dai, Hongsheng Li, and Ceyuan Yang. CameraCtrl: Enabling camera control for text-to-video generation. *arXiv preprint arXiv:2404.02101*, 2024. URL https://arxiv.org/abs/2404.02101.

Siqiao Huang, Jialong Wu, Qixing Zhou, Shangchen Miao, and Mingsheng Long. Vid2world: Crafting video diffusion models to interactive world models. In *International Conference on Learning Representations*, 2026. URL https://arxiv.org/abs/2505.14357.

InSpatio Team, Donghui Shen, Guofeng Zhang, Haomin Liu, Haoyu Ji, Hujun Bao, Hongjia Zhai, Jialin Liu, Jing Guo, Nan Wang, Siji Pan, Weihong Pan, Weijian Xie, Xianbin Liu, Xiaojun Xiang, Xiaoyu Zhang, Xinyu Chen, Yifu Wang, Yipeng Chen, Zhenzhou Fan, Zhewen Le, Zhichao Ye, and Ziqiang Zhao. INSPATIO-WORLD: A real-time 4d world simulator via spatiotemporal autoregressive modeling. *arXiv preprint arXiv:2604.07209*, 2026. URL https://arxiv.org/abs/2604.07209.

Ruilong Li, Brent Yi, Junchen Liu, Hang Gao, Yi Ma, and Angjoo Kanazawa. Cameras as relative positional encoding. In *Advances in Neural Information Processing Systems 38*, 2025a. URL https://proceedings.neurips.cc/paper_files/paper/2025/hash/17a7075094632c88cccdd86270ad715b-Abstract-Conference.html.

Runjia Li, Philip Torr, Andrea Vedaldi, and Tomas Jakab. VMem: Consistent interactive video scene generation with surfel-indexed view memory. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 25690–25699, 2025b. URL https://openaccess.thecvf.com/content/ICCV2025/html/Li_VMem_Consistent_Interactive_Video_Scene_Generation_with_Surfel-Indexed_View_Memory_ICCV_2025_paper.html.

Haotong Lin, Sili Chen, Junhao Liew, Donny Y. Chen, Zhenyu Li, Guang Shi, Jiashi Feng, and Bingyi Kang. Depth anything 3: Recovering the visual space from any views. *arXiv preprint arXiv:2511.10647*, 2025. URL https://arxiv.org/abs/2511.10647.

Ryan Po, Yotam Nitzan, Richard Zhang, Berlin Chen, Tri Dao, Eli Shechtman, Gordon Wetzstein, and Xun Huang. Long-context state-space video world models. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2025. URL https://arxiv.org/abs/2505.20171.

Xuanchi Ren, Tianchang Shen, Jiahui Huang, Huan Ling, Yifan Lu, Merlin Nimier-David, Thomas Müller, Alexander Keller, Sanja Fidler, and Jun Gao. GEN3C: 3D-informed world-consistent video generation with precise camera control. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.

Tianchang Shen, Sherwin Bahmani, Kai He, Sangeetha Grama Srinivasan, Tianshi Cao, Jiawei Ren, Ruilong Li, Zian Wang, Nicholas Sharp, Zan Gojcic, Sanja Fidler, Jiahui Huang, Huan Ling, Jun Gao, and Xuanchi Ren. Lyra 2.0: Explorable generative 3d worlds. *arXiv preprint arXiv:2604.13036*, 2026. URL https://arxiv.org/abs/2604.13036.

Xuelun Shen, Zhipeng Cai, Wei Yin, Matthias Müller, Zijun Li, Kaixuan Wang, Xiaozhi Chen, and Cheng Wang. GIM: Learning generalizable image matcher from internet videos. In *International Conference on Learning Representations*, 2024. URL https://proceedings.iclr.cc/paper_files/paper/2024/hash/38830ee2c33d56682385a00c8e7e9b85-Abstract-Conference.html.

Kiwhan Song, Boyuan Chen, Max Simchowitz, Yilun Du, Russ Tedrake, and Vincent Sitzmann. History-guided video diffusion. *arXiv preprint arXiv:2502.06764*, 2025. URL https://arxiv.org/abs/2502.06764.

Wenqiang Sun, Haiyu Zhang, Haoyuan Wang, Junta Wu, Zehan Wang, Zhenwei Wang, Yunhong Wang, Jun Zhang, Tengfei Wang, and Chunchao Guo. Worldplay: Towards long-term geometric consistency for real-time interactive world modeling. *arXiv preprint arXiv:2512.14614*, 2025. URL https://arxiv.org/abs/2512.14614.

Team Wan, Ang Wang, Baole Ai, Bin Wen, Chaojie Mao, Chen-Wei Xie, Di Chen, Feiwu Yu, Haiming Zhao, Jianxiao Yang, Jianyuan Zeng, Jiayu Wang, Jingfeng Zhang, Jingren Zhou, Jinkai Wang, Jixuan Chen, Kai Zhu, Kang Zhao, Keyu Yan, Lianghua Huang, Mengyang Feng, Ningyi Zhang, Pandeng Li, Pingyu Wu, Ruihang Chu, Ruili Feng, Shiwei Zhang, Siyang Sun, Tao Fang, Tianxing Wang, Tianyi Gui, Tingyu Weng, Tong Shen, Wei Lin, Wei Wang, Wei Wang, Wenmeng Zhou, Wente Wang, Wenting Shen, Wenyuan Yu, Xianzhong Shi, Xiaoming Huang, Xin Xu, Yan Kou, Yangyu Lv, Yifei Li, Yijing Liu, Yiming Wang, Yingya Zhang, Yitong Huang, Yong Li, You Wu, Yu Liu, Yulin Pan, Yun Zheng, Yuntao Hong, Yupeng Shi, Yutong Feng, Zeyinzi Jiang, Zhen Han, Zhi-Fan Wu, and Ziyu Liu. Wan: Open and advanced large-scale video generative models. *arXiv preprint arXiv:2503.20314*, 2025. URL https://arxiv.org/abs/2503.20314.

Jiahao Wang, Yufeng Yuan, Rujie Zheng, Youtian Lin, Jian Gao, Lin-Zhuo Chen, Yajie Bao, Yi Zhang, Chang Zeng, Yanxi Zhou, Xiao-Xiao Long, Hao Zhu, Zhaoxiang Zhang, Xun Cao, and Yao Yao. SpatialVID: A large-scale video dataset with spatial annotations. *arXiv preprint arXiv:2509.09676*, 2025a. URL https://arxiv.org/abs/2509.09676.

Qianqian Wang, Yifei Zhang, Aleksander Holynski, Alexei A. Efros, and Angjoo Kanazawa. Continuous 3d perception model with persistent state. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 10510–10522, 2025b. URL https://openaccess.thecvf.com/content/CVPR2025/html/Wang_Continuous_3D_Perception_Model_with_Persistent_State_CVPR_2025_paper.html.

Weijie Wang, Haoyu Zhao, Yifan Yang, Feng Chen, Zeyu Zhang, Yefei He, Zicheng Duan, Donny Y. Chen, Yuqing Yang, and Bohan Zhuang. Latent spatial memory for video world models. *arXiv preprint arXiv:2606.09828*, 2026a. URL https://arxiv.org/abs/2606.09828.

Zun Wang, Han Lin, Jaehong Yoon, Jaemin Cho, Yue Zhang, and Mohit Bansal. AnchorWeave: World-consistent video generation with retrieved local spatial memories. *arXiv preprint arXiv:2602.14941*, 2026b. URL https://arxiv.org/abs/2602.14941.

Zhengxuan Wei, Xu Guo, Xinghui Li, Xunzhi Xiang, Min Wei, Yiran Zhu, Qiulin Wang, Xintao Wang, Pengfei Wan, Xiangwang Hou, and Qi Fan. Geometry-aware implicit memory for video world models. *arXiv preprint arXiv:2606.02436*, 2026. URL https://arxiv.org/abs/2606.02436.

Ruiqi Wu, Xuanhua He, Meng Cheng, Tianyu Yang, Yong Zhang, Zhuoliang Kang, Xunliang Cai, Xiaoming Wei, Chunle Guo, Chongyi Li, and Ming-Ming Cheng. Infinite-world: Scaling interactive world models to 1000-frame horizons via pose-free hierarchical memory. *arXiv preprint arXiv:2602.02393*, 2026a. URL https://arxiv.org/abs/2602.02393.

Tong Wu, Shuai Yang, Ryan Po, Yinghao Xu, Ziwei Liu, Dahua Lin, and Gordon Wetzstein. Video world models with long-term spatial memory. *arXiv preprint arXiv:2506.05284*, 2025. URL https://arxiv.org/abs/2506.05284.

Xindi Wu, Sven Elflein, James Lucas, Olga Russakovsky, Laura Leal-Taixé, Despoina Paschalidou, Jonathan Lorraine, and Aljoša Ošep. Addressable memory for video world models. *arXiv preprint arXiv:2608.07408*, 2026b. URL https://arxiv.org/abs/2608.07408.

Zeqi Xiao, Yushi Lan, Yifan Zhou, Wenqi Ouyang, Shuai Yang, Yanhong Zeng, and Xingang Pan. WORLDMEM: Long-term consistent world simulation with memory. *arXiv preprint arXiv:2504.12369*, 2025. URL https://arxiv.org/abs/2504.12369.

Jiacong Xu, Hanwen Jiang, Zhixin Shu, Kalyan Sunkavalli, Vishal M. Patel, and Yiqun Mei. Wonder: Video world model done better. *arXiv preprint arXiv:2607.26037*, 2026a. URL https://arxiv.org/abs/2607.26037.

Tianxing Xu, Zixuan Wang, Guangyuan Wang, Li Hu, Zhongyi Zhang, Peng Zhang, Bang Zhang, and Song-Hai Zhang. UCM: Unified modeling of camera control and memory with time-aware positional encoding warping for world models. *arXiv preprint arXiv:2602.22960*, 2026b. URL https://arxiv.org/abs/2602.22960.

Zhufeng Xu, Xuan Gao, Bailin Deng, Yikang Ding, Xiaoqiang Liu, Haoxian Zhang, Pengfei Wan, Hongbo Fu, and Lin Gao. CameraSquad: Achieving content consistency in parallel multi-trajectory camera-controlled video generation. In *SIGGRAPH 2026 Conference Papers*. Association for Computing Machinery, 2026c. doi: 10.1145/3799902.3811048. URL https://doi.org/10.1145/3799902.3811048.

Jiwen Yu, Jianhong Bai, Yiran Qin, Quande Liu, Xintao Wang, Pengfei Wan, Di Zhang, and Xihui Liu. Context as memory: Scene-consistent interactive long video generation with memory retrieval. In *Proceedings of the SIGGRAPH Asia 2025 Conference Papers*, pp. 19:1–19:11, 2025a. doi: 10.1145/3757377.3763833. URL https://doi.org/10.1145/3757377.3763833.

Mark Yu, Wenbo Hu, Jinbo Xing, and Ying Shan. TrajectoryCrafter: Redirecting camera trajectory for monocular videos via diffusion models. *arXiv preprint arXiv:2503.05638*, 2025b. URL https://arxiv.org/abs/2503.05638.

Wei Yu, Runjia Qian, Yumeng Li, Liquan Wang, Songheng Yin, Sri Siddarth Chakaravarthy P, Dennis Anthony, Yang Ye, Yidi Li, Weiwei Wan, and Animesh Garg. MosaicMem: Hybrid spatial memory for controllable video world models. *arXiv preprint arXiv:2603.17117*, 2026. URL https://arxiv.org/abs/2603.17117.

Cheng Zhang, Boying Li, Meng Wei, Yan-Pei Cao, Camilo Cruz Gambardella, Dinh Phung, and Jianfei Cai. Unified camera positional encoding for controlled video generation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 38027–38037, 2026. URL https://openaccess.thecvf.com/content/CVPR2026/html/Zhang_Unified_Camera_Positional_Encoding_for_Controlled_Video_Generation_CVPR_2026_paper.html.

Lvmin Zhang, Shengqu Cai, Muyang Li, Gordon Wetzstein, and Maneesh Agrawala. Frame context packing and drift prevention in next-frame-prediction video diffusion models. In *Advances in Neural Information Processing Systems 38*, 2025. URL https://proceedings.neurips.cc/paper_files/paper/2025/hash/2bde8fef08f7ebe42b584266cbcfc909-Abstract-Conference.html.

Yian Zhao, Ruochong Zheng, Hongcan Guo, Yu Yan, Jian Zhang, and Jie Chen. Miniworld: Democratizing the training of video world models from scratch. *arXiv preprint arXiv:2608.01127*, 2026. URL https://arxiv.org/abs/2608.01127.

Guangcong Zheng, Teng Li, Xianpan Zhou, and Xi Li. RealCam-Vid: High-resolution video dataset with dynamic scenes and metric-scale camera movements. *arXiv preprint arXiv:2504.08212*, 2025. URL https://arxiv.org/abs/2504.08212.

A Retrieval and Positional-Encoding Details

A.1 Spectral Regularization

Operator settings. After projection and head reshaping, tensors have layout $[B, L, N_{\text{head}}, D]$. Positional operators act independently on each head's D channels before scaled dot-product attention, without mixing heads or token rows. The output transform B_q^O precedes head concatenation and output projection. Each token's σ_h is shared across heads. We use $\lambda_c = 0.75$ and $\lambda_a = 0.25$, with widths measured in transported patch-grid units and depth differences computed on the aligned historical depth grid. These coefficients give $\sigma_h \in [0, 1]$, and the operator characterization covers $\sigma \in [0, 2]$.

Continuous-coordinate motivation. For fixed source and target cameras and fixed content, suppose only the continuous transported coordinate is perturbed by $\epsilon \sim \mathcal{N}(0, \sigma^2 I_2)$. The Gaussian characteristic function gives

$$\mathbb{E}[R_k(a + \epsilon_a)] = e^{-\sigma^2 \omega_k^2/2} R_k(a), \qquad \mathbb{E}[B_h^{KV}(a + \epsilon) K^{\text{pre}}] = D_\sigma B_h^{KV}(a) K^{\text{pre}}. \tag{11}$$

The same conditional identity holds for V^{pre}. It follows blockwise because the camera block is fixed and the two spatial blocks are axial Fourier pairs, as in integrated positional encoding (Barron et al., 2021).

Measured address error. The width is intended to be larger where a transported address is less reliable, which is testable without ground-truth geometry. For a held-out sample of 60 clips we pair a history frame with a later frame at separations of 2, 4 and 8 latent steps and run GIM-DKM (Shen et al., 2024; Edstedt et al., 2023) dense matching between the two real RGB frames, retaining correspondences above 0.9 confidence. The matched location of each patch centre is an estimate of where that surface actually appears in the queried view, obtained without any estimated depth or pose, so it is independent of the geometry that produced the transported coordinate. Comparing it against the DA3 projection over 41,661 retained patch centres, the median address error is 0.37 patch cells and σ_h increases with it (Spearman $\rho = 0.329$, Pearson $r = 0.383$, $p < 10^{-150}$).

Reference settings.

Candidate limits	Up to 16 source and 16 generated views per target, and the all-source inference option removes only the source cap.
Footprint score	$4H_p \times 4W_p$ grid, 16 samples per patch, and source-priority selection exhausts positive gains.
Local read	$M = H_pW_p$ original tokens per target frame, and frame-local adapter mask, distinct from native temporal attention.
Layer schedule	Interaction stride 1, and adapter width 3072, 24 heads, 128 channels per head.
Initialization	Pointer/global scalar gates and outer output projection start at zero, and adapter input projection starts at identity.

The following sections specify source-priority retrieval, camera ordering, token allocation, and positional encoding.

Three resolutions and two selections. Let the isomorphic patch grid be $H_p \times W_p$, after the shared DiT patch embedding, with $M = H_pW_p$ tokens per frame. Source RGB pixels, these patch cells, and the $4H_p \times 4W_p$ footprint scoring grid are distinct. Increasing the scoring resolution does not create additional attention tokens. The footprint routine ranks candidate *views*. A subsequent routine chooses original *tokens* from those views using center-cell coverage. Consequently the area covered by all patches in selected views is not necessarily the area supported by the final M tokens.

Candidate bank and ordering. The bank contains decoded historical observations and their aligned DA3 depth/pose/intrinsics, with an immutable-source versus generated-history flag. For each target frame, camera prefiltering uses $s_i = \theta_i/(15^\circ) + \|\mathbf{t}_i - \mathbf{t}_q\|/0.25$, where translation is in the scene-depth-normalized gauge. The nearest view is first, and subsequent views are ordered by $s_i - \alpha_i$, with stable observation order breaking ties. The default shortlist retains at most 16 immutable-source and 16 generated views (16 total when no source flag is supplied). The all-source inference option removes the former cap but keeps the generated-view cap. Compared retrieval variants share this setting and the upstream history sampling schedule, and the attention budget remains M independently of shortlist size. The shortlist is per target frame, while projection can batch its union across a target window.

View selection scores sampled geometric footprints, whereas positional addressing uses the projected centers of the selected tokens, and the two stages therefore consume different geometric primitives.

Projected footprint scoring. For each source patch, the routine unprojects its four grid corners using one depth value shared by the corners and the grid-scaled source intrinsics, then projects them into the target. A patch contributes only if all four corners have positive target-camera depth ($> 10^{-6}$). Sixteen bilinear sample locations inside the projected quadrilateral are rounded onto the $4H_p \times 4W_p$ grid, and in-bounds samples mark occupied cells. Their Boolean union defines the projected-occupancy mask F_i, counting cross-view overlaps once. Table 13 compares this retrieval strategy with center-hit and causal z-buffer controls.

Objective and implemented greedy policy. For a fixed bank, define

$$C(S) = \frac{|\bigcup_{i \in S} F_i|}{16H_pW_p}, \qquad \Delta(i \mid S) = \frac{|F_i \setminus \bigcup_{j \in S} F_j|}{16H_pW_p}. \tag{12}$$

This fixed-set union objective is monotone and submodular. The implemented policy adds camera prefiltering, a seeded first view, source priority, and separate token allocation on top of it.

The main-text retrieval-to-read procedure specifies the source-priority branch. Equal marginal gains retain shortlist order, and accepting the seed also updates U with its footprint mask. Selection follows the source-priority policy in the main text and exhausts positive coverage gains.

Token ownership, fillers, and PE. The budget allocator projects source-token centers and quantizes them by $\lfloor u + 0.5 \rfloor, \lfloor v + 0.5 \rfloor$. In a collision, the first token in selected-view/source-cell order receives coverage priority. Remaining capacity is filled with unused original tokens, which may be redundant, out of bounds, or behind the target camera. Such fillers remain valid attention rows but do not count as target-visible coverage. Invalid padding rows are excluded. A sign-preserving 10^{-6} denominator floor avoids division by zero, and out-of-frame positions are not clipped into the target image. Each

selected token retains source frame ID, source cell ID, its continuous projected center, and reliability metadata. Its PE uses the continuous projected center.

Tensor layout. For tensors $[B, L, N_h, D]$, the first $D/2$ channels are grouped into four-channel projective vectors. In each following $D/4$ block, the two halves form $D/8$ Fourier pairs, rather than adjacent even/odd channels. The attenuation vector is repeated over these two halves and broadcast over batch/head dimensions. It is applied after the projective/spatial transform, only to pointer K/V spatial blocks. Equation 11 follows for each sine and cosine component separately.

A.2 Projection Convention and Layer Sequence

Let P_q and P_h denote the lifted, intrinsics-aware world-to-camera matrices of the target and historical token's source view in a shared scene gauge. We adopt the PRoPE head partition $D/2{+}D/4{+}D/4$ (Li et al., 2025a). The first block contains repeated four-dimensional camera groups, and the remaining blocks contain horizontal and vertical rotary pairs. Define

$$\begin{aligned} B_q^Q(u,v) &= \mathrm{diag}\big(C(P_q^\top), R_x(u), R_y(v)\big)\,, \\ B_h^{KV}(u,v) &= \mathrm{diag}\big(C(P_h^{-1}), R_x(u), R_y(v)\big)\,, \\ B_q^O(u,v) &= \mathrm{diag}\big(C(P_q), R_x(u)^{-1}, R_y(v)^{-1}\big)\,. \end{aligned} \tag{13}$$

Here $C(P)$ applies P independently to each four-channel group. Memory K/V retain their source-camera basis while their spatial coordinates are projected into the target view. The camera blocks therefore contribute $P_q P_h^{-1}$ to query–key interactions and value transport into the target basis.

Geometry and coordinate units. Forward depth differences in Equation 8 are zero-padded at the right and bottom boundaries. Absent explicit observation times, history receives equally spaced times in sequence order. $\mathrm{round}_{\mathrm{even}}$ in Equation 9 is `torch.round`, not floor quantization: $-1.5, -0.5, 0.5, 1.5$ map to $-2, 0, 0, 2$. Historical depth is bilinearly resized to the source patch grid with `align_corners=False`, and intrinsics are row-scaled to that grid before unprojection, associating a resized depth sample with each patch center. Source and target camera-to-world translations and depth must use a common scene scale. Scene normalization translates the reference origin and divides positions and depths by the same scene-depth scale, and it does not independently rescale each camera. Target pixel coordinates $(u_{\mathrm{pix}}, v_{\mathrm{pix}})$ become $u = u_{\mathrm{pix}} W_p / W_{\mathrm{img}} - 1/2$ and $v = v_{\mathrm{pix}} H_p / H_{\mathrm{img}} - 1/2$.

Pointer-shadow update at layer ℓ. Let x^ℓ denote the main-stream features and p^ℓ the packed historical features. The shared patch embedding is applied once at input, and subsequent layers operate on the updated hidden states. For the pointer-shadow branch, the code executes the following sequence:

1. Use the current backbone block's normalization, Q/K/V projections, and timestep modulation to process p^ℓ at timestep zero. Pointer Q/K receive native DiT RoPE at their original source-grid coordinates, and target-view transport is applied at the subsequent memory read.
2. Pointer queries attend to pointer tokens associated with the same target frame, without reading noisy target features or other target-frame groups. The same block's text attention and feed-forward weights produce $\hat{p}^{\ell+1}$. A learned update gate yields $p^{\ell+1} = p^\ell + \tanh(g_\ell)(\hat{p}^{\ell+1} - p^\ell)$.
3. Run the ordinary target backbone block on x^ℓ. Its native target-to-target temporal attention is retained. In this shadow branch, this ordinary pass is distinct from appending pointer tokens directly to one joint backbone sequence.
4. At configured adapter interaction layers, the updated target features read the refreshed pointer features through the frame-local STaPE/PRoPE operation in Equation 7. CUT3R has a separate softmax, and its output is gated after attention, before the combined conditioning update is projected back into the main stream.

The reference pointer-shadow configuration uses `interact_every=1`, so refresh and adapter read occur at every backbone layer. More generally, stride k selects zero-based blocks with $\ell \bmod k =$

$k-1$. Each adapter has width 3072, 24 heads of width 128, and its own parameters. Its input projection is initialized to the identity, and Q/K/V and the inner output projection copy the corresponding DiT attention weights at initialization, without subsequent weight tying. Adapter LayerNorms remain separate from the native block's normalization.

Both the pointer update parameter g_ℓ and the global-read parameter β_ℓ are scalar, learned, and initialized to zero. If $o_{L,\ell}$ is the local read after the inverse output-coordinate transform and $o_{G,\ell}$ is the separately normalized global read, the target update is

$$x^{\ell+1}_{\text{target}} = \hat{x}^{\ell+1}_{\text{target}} + B_\ell W_{O,\ell}\big(o_{L,\ell} + \tanh(\beta_\ell) o_{G,\ell}\big). \tag{14}$$

Here $\hat{x}^{\ell+1}$ is the ordinary backbone output and B_ℓ is the zero-initialized outer output projection. There is no additional scalar multiplying the local read in this `zero_output` mode. The source/TiER prefix is not modified by this adapter residual. Each token's σ_h is broadcast over attention heads.

B Training and Evaluation Details

B.1 Noise Schedule and Prediction Convention

The dense reference trainer samples an index uniformly from $N = 1000$ scheduler nodes. Before numerical timestep casting, node j is $u_j = 1 - j/N$, $s_j = \rho u_j/[1 + (\rho - 1)u_j]$, and $t_j = 1000 s_j$, for $j = 0, \ldots, N-1$. The wrapper's reference shift is $\rho = 5$. The scheduler weights in Equation 10 are

$$y_j = \exp\left[-2\left(\frac{t_j - N/2}{N}\right)^2\right], \qquad w_j = \frac{N(y_j - \min_i y_i)}{\sum_i (y_i - \min_r y_r)}. \tag{15}$$

The squared latent-space velocity prediction error is averaged over latent elements before applying this weight. The ordinary training path casts the model timestep to BF16, and noise construction and loss weighting use the nearest stored scheduler node to that supplied timestep. This numerical lookup does not redefine the velocity as a derivative with respect to the timestep index. The displayed rollouts use 50 Euler steps with the shifted schedule and a final update to zero noise, and no stochastic SDE term is added.

Future donors. The future augmentation is training-only. With target start s and length $T = 20$ on the latent-aligned observation clock, the trainer checks every donor index $j > s + T + g$. We use $g = 40$, donor sampling stride 15, at most 16 donor observations, and attempt probability 0.70. These are observation-index units, not seconds, and frame rate and temporal subsampling determine their physical duration. The reader selects available donors beyond this guard and supplies their indices explicitly, and the trainer does not assume a uniform continuous-time distribution. Donors are sampled only when eligible observations are available. RGB, depth, pose, and intrinsics retain the same true donor index. Only the timestamps used to compute memory temporal lag are relabeled, and target RGB/cameras are unchanged. No future donor is available in causal rollout inference.

Revisit and fast-motion augmentation. Synthetic revisit perturbs multiple selected memory views using a shared camera-local delta, leaving target RGB and target camera commands unchanged. Revisitation is attempted with probability 0.50, with perturbations up to 45° yaw, 0.20 scene-depth units laterally, and 0.12 units forward. A joint-bank option permits future donors to receive revisit perturbations as well, and these two augmentations are not inherently mutually exclusive. Fast-motion augmentation uses temporally subsampled targets with synchronously indexed RGB and geometry, not a change of playback FPS alone.

Future donors provide additional visual support during training, and inference uses causal observations. Training/test separation is defined at source-video level and includes derived clips, donor pools, and geometry caches.

Table 7 isolates each augmentation by retraining the complete system with it removed.

B.2 Experimental Settings and Comparison Protocols

Training configuration. We train the backbone of Section 3.3 at increasing resolution, from 256×448 to 352×640, with a 20-latent-frame target window. Training uses a mixture of SpatialVID,

Table 7: Training-augmentation ablation. Each row retrains the complete system with the indicated augmentations and is evaluated on the same 100-case set under the same causal inference protocol. The causal-only variant isolates the contribution of training-time future-as-memory. Column definitions follow Table 5.

Configuration	Future-as-memory	Synthetic revisit	Fast motion	Mat. Pix.@5 (K) ↑	Inliers@5 (K) ↑	RGB MAE@5 ↓	Mean rot. (°) ↓	Trans. NRMSE ↓	CLIP-T ↑	CLIP-F ↑
All augmentations	On	On	On	7.153	6.480	0.109	1.16	0.204	30.089	99.13
No future-as-memory (causal only)	Off	On	On	6.851	5.918	0.134	1.53	0.288	28.172	97.12
No synthetic revisit	On	Off	On	6.929	6.020	0.147	1.33	0.273	28.090	97.03
No fast motion	On	On	Off	7.011	6.220	0.119	1.78	0.312	29.100	96.90
No augmentation	Off	Off	Off	6.629	5.644	0.162	1.90	0.338	27.936	96.02

RealCam-Vid, ReCamMaster, and Context-as-Memory. Cached DA3 (Lin et al., 2025) pose, intrinsics, and depth provide geometry, and frozen CUT3R with MeMix (Wang et al., 2025b; Dong et al., 2026) provides recurrent context. Internal controls hold initialization, data, optimization budget, geometry, and augmentations fixed except for the factor under study. We train for 40,000 optimization steps using AdamW with learning rate 10^{-6} for the backbone and memory-injection modules, weight decay 10^{-4}, $\beta = (0.9, 0.999)$, and $\epsilon = 10^{-8}$. Training uses eight GPUs with one sample per GPU and four gradient accumulation steps, giving an effective batch size of 32. The DiT backbone and memory-injection modules are optimized jointly, while the VAE, text encoder, and recurrent geometry encoder remain frozen.

Tasks and inference settings. Camera replay follows trajectories estimated from the corresponding source videos. Y-traj. observes the left and right sides before moving backward at the initial heading, producing 240 new RGB frames plus the source frame. Internal paired comparisons share the observed prefix, prompt, requested cameras, seed list, resolution, sampler, and horizon. Inference donors consist of observed source frames and completed generated windows. Generated endpoints are retained as generated. Source-video-disjoint splits separate derived clips, donor pools, and geometry caches.

STaPE controls. Table 5 holds the observation pool, cameras, footprint retrieval, global branch, and local read budget fixed. Latent/value warping resamples historical features onto the target grid. PE warp uses continuous coordinates in every spatial band and disables spectral attenuation. The two intermediate controls enable multi-precision coordinates or spectral attenuation separately, and the full configuration enables both. Attenuation modulates both memory keys and values.

Memory controls. Table 6 independently enables GRM and LPM. GRM supplies frozen chronological recurrent context, and LPM supplies geometry-indexed patches with full STaPE. Source, text, and camera conditioning are retained in all four configurations. The local read budget is fixed whenever LPM is enabled.

Metrics and aggregation. The camera-control table reports mean and endpoint rotation errors, translation NRMSE, translation-direction error, CLIP-T, CLIP-F, FID, and FVD. DA3-derived pose scores quantify agreement with the requested trajectory in the estimated scene coordinate system. WorldScore component scaling and composite weighting are given in the table captions. For per-case metrics, frame or frame-pair measurements are averaged within each case, followed by an equal-weight average over cases. Inlier fractions are computed per pair before this aggregation. For revisit correspondences, each video pairs frames 76–80 and 156–160 with frames 236–240, using zero-based one-to-one indices, and GIM-DKM (Shen et al., 2024; Edstedt et al., 2023) samples 15,000 correspondences per pair after a common 240×416 resize and padding, and matches above 0.1 confidence are counted (Mat. Pix.). Geometric inliers are estimated with a fundamental matrix using OpenCV USAC_MAGSAC, a 1-pixel threshold in the unpadded 240×416 coordinates, confidence 0.999999, and at most 10,000 iterations. Inliers are counted only among matches above 0.1 confidence, and RGB MAE is evaluated at these accepted correspondences using bilinear sampling of RGB values in $[0, 1]$. FID and FVD are computed over the evaluation set as distribution-level metrics. Motion magnitude measures the median displacement of estimated optical flow rather than camera accuracy, and flow consistency measures forward–backward optical-flow agreement. On the Y-traj. set, all six methods use the full shared case set and the same evaluation protocol. Appendix B.3 defines the counting units and aggregation details.

External comparisons. The experiments compare GLIMPSE with the state-of-the-art systems that have public implementations, namely UCM, SPMem, GEN3C, VMem, and ReWorld. We follow the inference recipes provided in their public repositories, adapting input and camera interfaces to the evaluation tasks. The system-level tables compare complete generation pipelines, and the internal ablations isolate changes within GLIMPSE's shared backbone and training recipe.

VMem configuration. VMem uses a single source image, native 576×576 generation, 50 sampling steps, guidance 2.0 (minimum 1.2), four context and four target frames per model call, and 400 surfel iterations. The source raster is resized with matching intrinsics scaling, and camera axes are converted at the adapter boundary. WorldScore uses frames 0 and 80 as anchors and the original source image as the style reference.

B.3 Evidence Records and Aggregation

Diagnostic scope. Projected footprint occupancy measures image-plane support. Target-depth agreement additionally tests positive depth, image bounds, and agreement with a reference surface using reference target depth for evaluation. The fixed budget bounds local attention read length, and geometry estimation, retrieval, transfer, denoising, and storage account for the remaining system costs.

B.4 STaPE Mechanism Validation

Key and value modulation. We compare modulation of keys, values, and both, retaining footprint retrieval, global memory, and the local read budget. Multi-precision coordinates remain enabled. To separate frequency-selective modulation from scalar suppression, the scalar control replaces the frequency-dependent multipliers by their per-token mean on the same spatial components of K and V, providing a magnitude-control baseline.

Table 8: Inference-time attenuation-path comparison. None disables attenuation.

Modulation	Mat. Pix.@5 (K)↑	Inliers@5 (K)↑	RGB MAE @5↓
None	6.567	5.707	0.122
K only	6.650	5.985	0.114
V only	6.501	6.199	0.125
K + V (STaPE)	7.153	6.480	0.109
Matched scalar on K + V	6.864	6.272	0.119

Table 9: Reliability-signal comparison. Coordinate precision and joint K/V modulation are fixed. Depth + temporal lag uses full STaPE.

Width signal	Mat. Pix.@5 (K)↑	RGB MAE @5↓
Depth only	6.799	0.126
Temporal lag only	6.723	0.119
Depth + temporal lag	7.153	0.109
Permuted depth + temporal lag	6.692	0.194

Joint K/V modulation achieves the highest correspondence counts. Compared with the matched scalar control, STaPE increases geometric inliers by 3.32% and reduces RGB MAE by 8.40%, supporting frequency-selective attenuation.

Depth and temporal-lag signals. The reliability diagnostic compares depth-only, temporal-lag-only, and combined widths, and a control that permutes the combined widths among retrieved tokens within each query, preserving their distribution. Coefficients retain their default values, $\lambda_c = 0.75$ and $\lambda_a = 0.25$, when the other term is removed. The combined row uses full STaPE.

Combining depth and temporal lag improves both metrics over either signal alone. Relative to permuted widths, the combined signal increases matched pixels by 6.89% and reduces RGB error by 43.81%, supporting token-specific modulation.

Sensitivity to address perturbations. We compare continuous PE transport and full STaPE using identical cases, generation seeds, and retrieved tokens. Independent zero-mean Gaussian offsets with standard deviation s are added to the two projected spatial coordinates before positional encoding and, for STaPE, before quantization. Both variants receive identical offsets, held fixed throughout denoising. Retrieval, content, and STaPE's depth- and temporal-lag-conditioned widths remain unchanged, isolating sensitivity to address errors.

Table 10: Address perturbations with standard deviation s in patch-grid units.

s	Addressing	Mat. Pix.@5 (K)↑	Inliers@5 (K)↑	RGB MAE@5↓
0	PE warp	6.407	5.690	0.135
	STaPE	7.153	6.480	0.109
0.25	PE warp	6.043	5.205	0.159
	STaPE	7.100	6.311	0.120
0.50	PE warp	6.019	4.897	0.183
	STaPE	6.894	6.242	0.139
1.00	PE warp	5.229	4.288	0.216
	STaPE	6.245	5.776	0.145

STaPE retains more matched pixels and geometric inliers, with lower RGB MAE, at every tested nonzero perturbation level. At $s = 1$, it yields 19.4% more matched pixels and 34.7% more inliers than PE warp, while reducing RGB MAE by 32.9%, supporting robust memory reads under perturbed positional addresses.

B.5 Retrieval Support and Exploration Cost

Retrieval support. Tables 11 and 12 summarize six rollouts from five sources, with 220 tokens per target frame. Return denotes an additional corrective window. Unique cells count distinct in-bounds cells relative to the target grid, and repeats count excess tokens in occupied cells relative to the read budget. Neither measures visibility.

Table 11: Retrieval support (%). Sources are equally weighted, and brackets give 95% source-cluster bootstrap intervals. Source share denotes tokens from original observations.

Window	In bounds [95% CI]	Unique cells	Repeats	Source share
1	61.2 [45.9, 76.4]	59.9	1.2	100.0
2	68.3 [62.1, 73.2]	60.0	8.3	81.4
3	75.0 [64.2, 86.6]	72.8	2.1	79.1
Return	95.2 [90.1, 98.6]	94.6	0.6	87.4

Table 12: Median component times over six rollouts. DA3 inputs include source and generated observations. Other pipeline stages are excluded.

Window	DA3 observations	DA3 forward (s)	Pointer cache (s)
1	41	1.68	3.61
2	61	2.72	2.37
3	81	4.14	3.92
Return	101	5.47	4.23

For a single 241-frame revisit rollout, the third window contains 20 latent queries with 220 retrieved tokens each. In-bounds tokens account for 95.41% of the read budget, distinct in-bounds cells cover 94.27% of the target grid, and repeated in-bounds projections account for 1.14% of the read budget.

Visibility diagnostic. The comparison fixes donors, cameras, checkpoint, seeds, and token budget. Valid support counts positive-depth, in-bounds tokens relative to the read budget. Depth consistency is the fraction of reference-valid projections with $|z_{\mathrm{proj}} - z_{\mathrm{ref}}|/z_{\mathrm{ref}} \leq 0.1$. The causal z-buffer uses observed history for selection. Reference target depth is used only to evaluate depth consistency.

Horizon profiling. Table 14 reports cumulative denoising and DA3 forward times for 81-, 241-, and 721-frame rollouts at 352×640, 50 denoising steps per window, and CFG 5.

Table 13: Retrieval comparison. Support columns are percentages, and @5 follows Table 4.

Selector	Valid (%)	Depth consistent (%)	Inliers@5 (K)
Footprint	92.09	37.00	6.923
Center-hit	90.18	31.59	6.030
Causal z-buffer	89.75	36.21	6.852

Table 14: Cumulative component times (seconds). Profiling includes diagnostic recording. DA3 forward times exclude model loading.

Frames	Denoising (s)	DA3 (s)
81	174.26	1.33
241	533.40	7.52
721	1580.41	55.92

C REPRODUCIBILITY STATEMENT

We have submitted the core code and visualization videos as supplementary materials. The full code and model weights will be publicly released upon acceptance.

AI use statement. We used AI tools to polish the manuscript's language and assist in searching for related literature.

D ADDITIONAL QUALITATIVE RESULTS

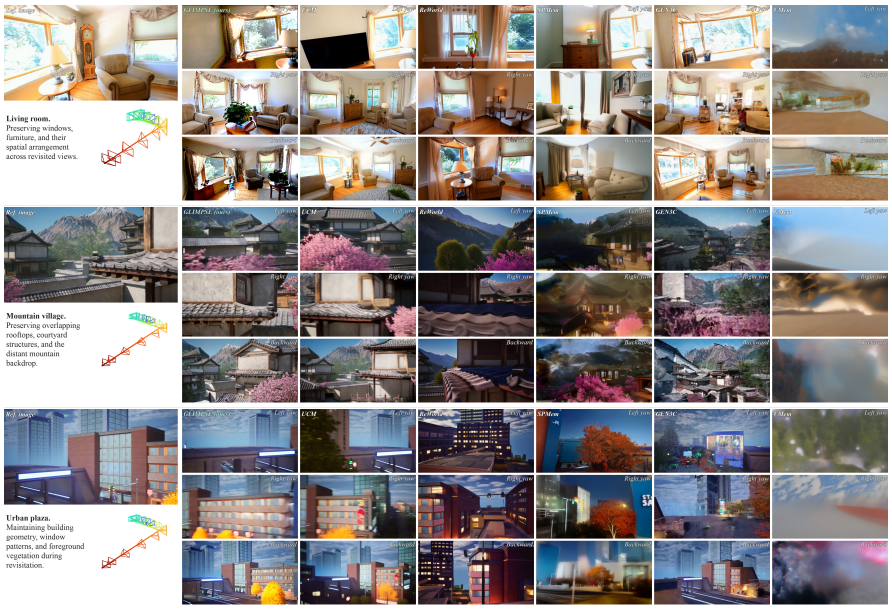


Figure 7: Additional Y-traj. comparisons in a living room, mountain village, and urban plaza, following the layout of Figure 5. Each scene shows left-yaw, right-yaw, and backward views across six methods.

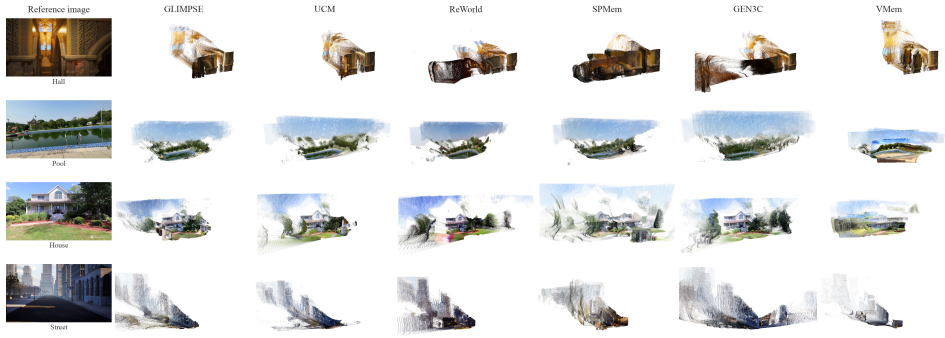


Figure 8: DA3 reconstructions of generated videos, colored by generated RGB. Methods share reconstruction settings, source-depth alignment, and a common rendering viewpoint and display scale per scene. We sample frames 8–80 at stride 8 for the hall and pool, and 16–224 at stride 16 for the house and street. The shared source anchors reconstruction but is not displayed.